

Workshop 6: statistics in R

Summary sheet

Two categorical variables

Assuming that:

- Observations independent
- Categories mutually exclusive

First make a **contingency table**:

```
cont_tab <- table(
  df$var1, df$var2)
```

Then use `chisq.test(cont_tab)`
if >5 observations per cell

or use `fisher.test(cont_tab)`
if sample size is small

Two continuous variables

Calculate **correlation** using the
`cor.test()` function.

Two common types: **pearson** and
spearman. Choose with the `method`
argument to `cor.test()`

Assumptions:

- Pearson: linear, minimal outliers
- Spearman: monotonic relationship

```
cor.test(df$var1, df$var2,
  method = "pearson")
```

One continuous, one categorical

Compare the means of 2 groups with a
t-test or of 3+ groups with an **ANOVA**

Assumptions:

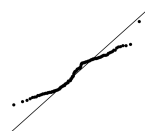
- Observations independent
- Observations approx. normal
- Variances of groups equal

```
t.test(var1~ var2, data = df)
aov(var1~ var2, data = df) %>%
  summary() ← for p value
OR
  TukeyHSD() ← for pairwise p values
```

If variances unequal:

Use `var.equal = FALSE` argument to
`t.test()` or `oneway.test()`
`oneway.test()` is the unequal variance version
of `aov()`

If normality violated:



QQ plot deviates from straight line
`shapiro.test()` significant

Use `wilcox.test()` instead of t-test and
`kruskal.test()` instead of ANOVA

Visualising simple stats with ggpubr

```
ggboxplot(data, var1, var2) +
  stat_compare_means()
ggscatter(data, var1, var2) +
  stat_cor()
```

Linear models

Assumptions:

- Linearity
- Residuals independent
- Residuals approx. normal
- Residuals have constant variance

Fit a linear model using:

```
lm(var1 ~ var2, data = df)
%>% summary()
```

where `var1` is continuous and `var2` can
be either continuous or discrete.

Test **differences relative to zero** using:

```
lm(var1 ~ 0 + var2, data = df)
%>% summary()
```

where **both** `var1` and `var2` are discrete.

Add **covariates** using:

```
lm(var1 ~ var2 + var3 + ...,
  data = df) %>% summary()
```

to try and control for confounding variables
in your analysis.

Best practices for stats in R

1. Always visualise your data first!
2. Check your assumptions
3. Run the appropriate statistical test
4. Interpret the results in plain language
5. Check that the results make sense with
your data visualisation & descriptive
statistics